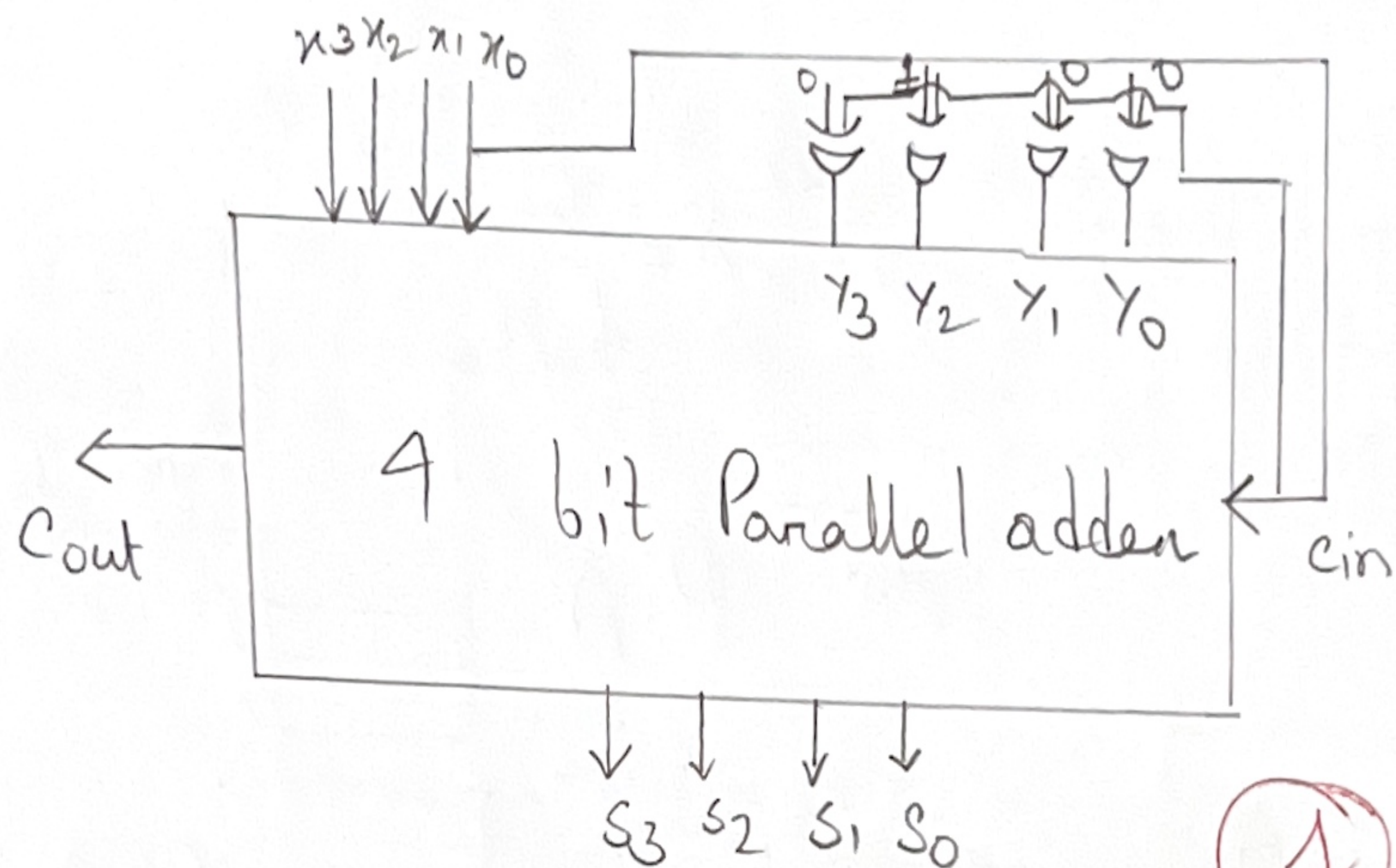
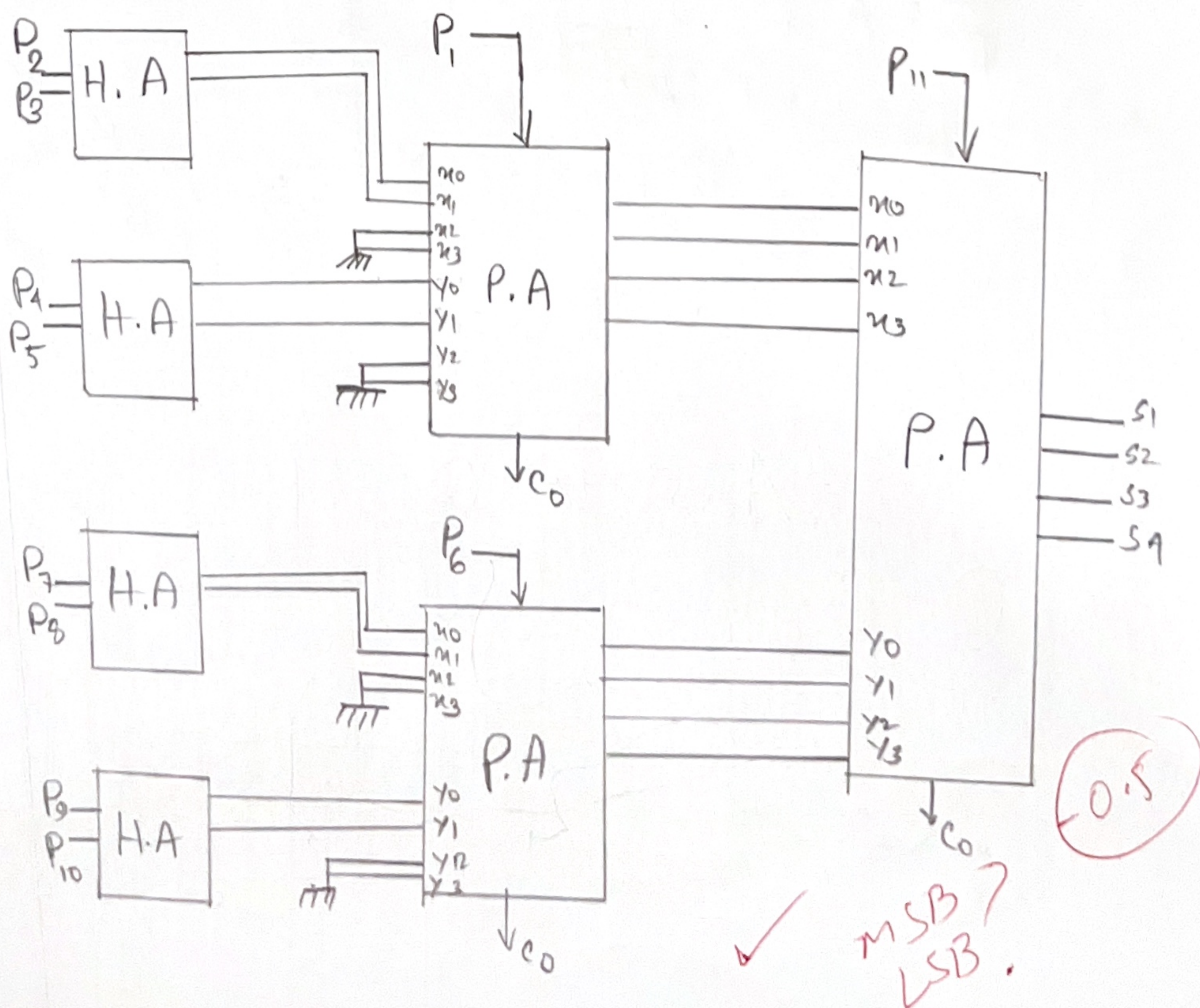


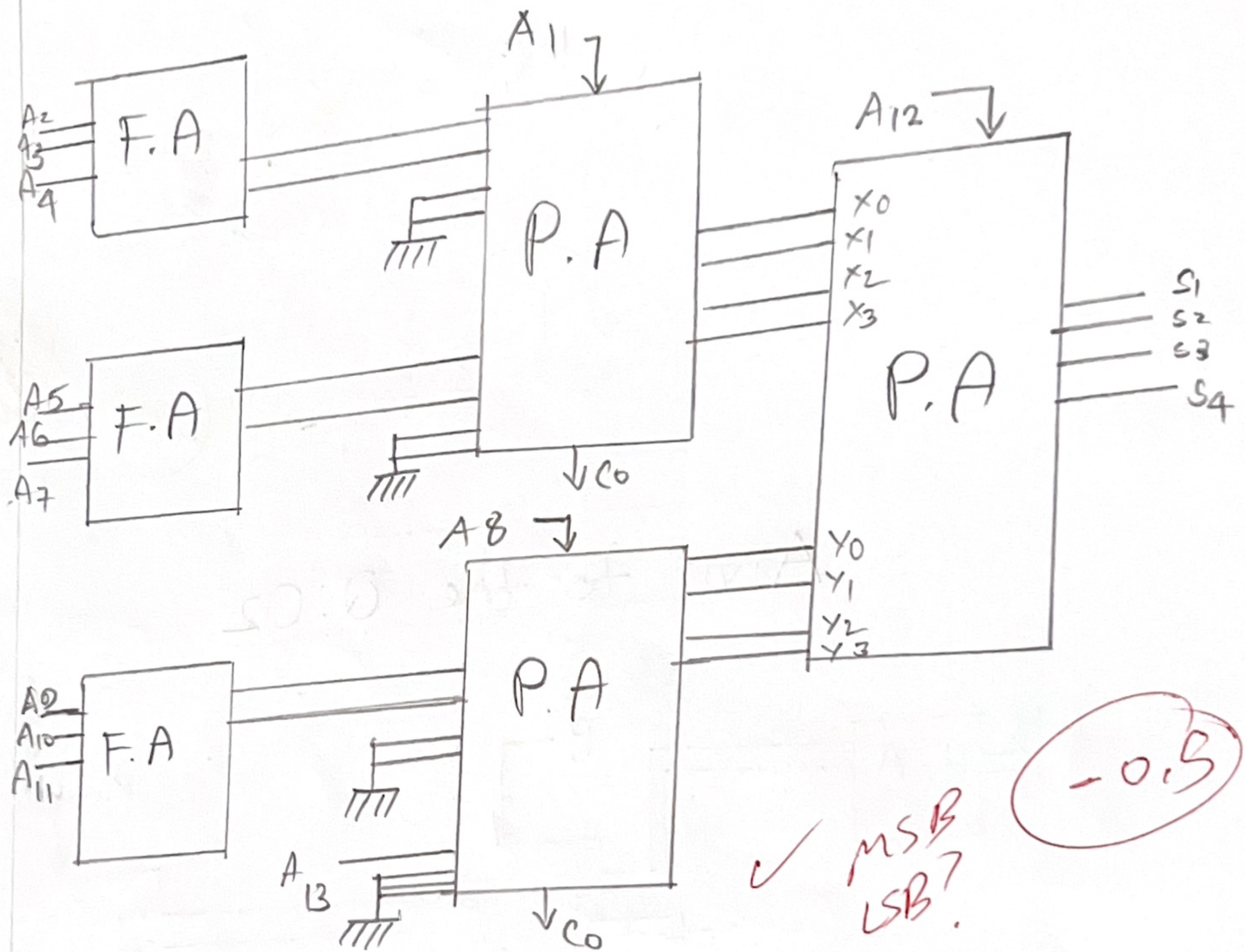
Ans: to the Q:01



Ans: to the Q:02

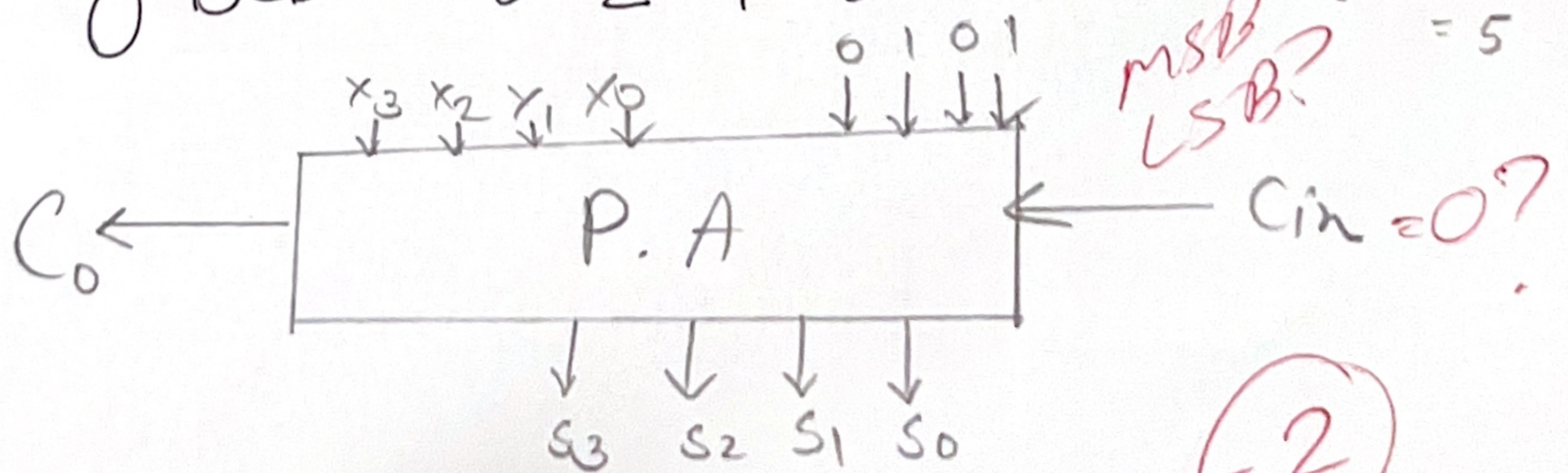


Ans: to Q: 03



Ans: to Q: 04

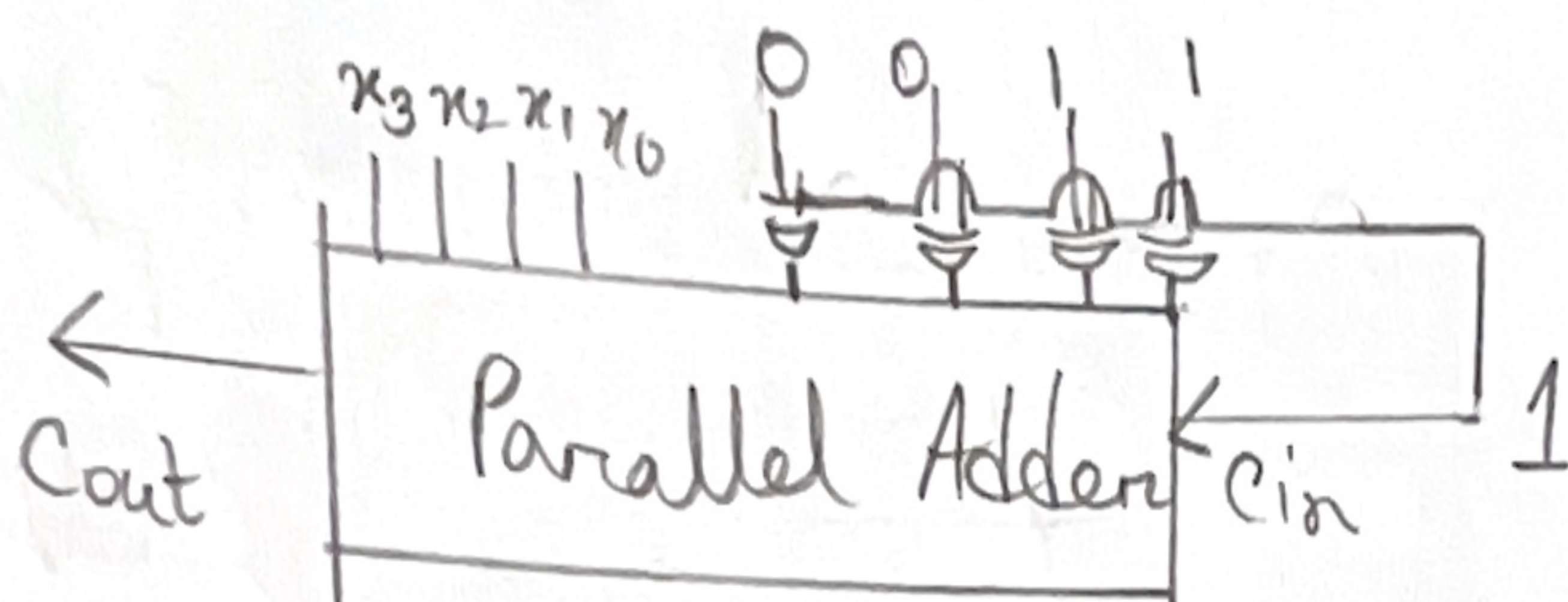
① By BCD: $x_3 x_2 x_1 x_0$



R.W

8	4	2	1
0	1	0	1
= 5			

② Excess: 3 $\div x_3 x_2 x_1 x_0$



D.W

$$\begin{array}{r} 8421 \\ \hline 0011 \end{array}$$

MSB
LSB?

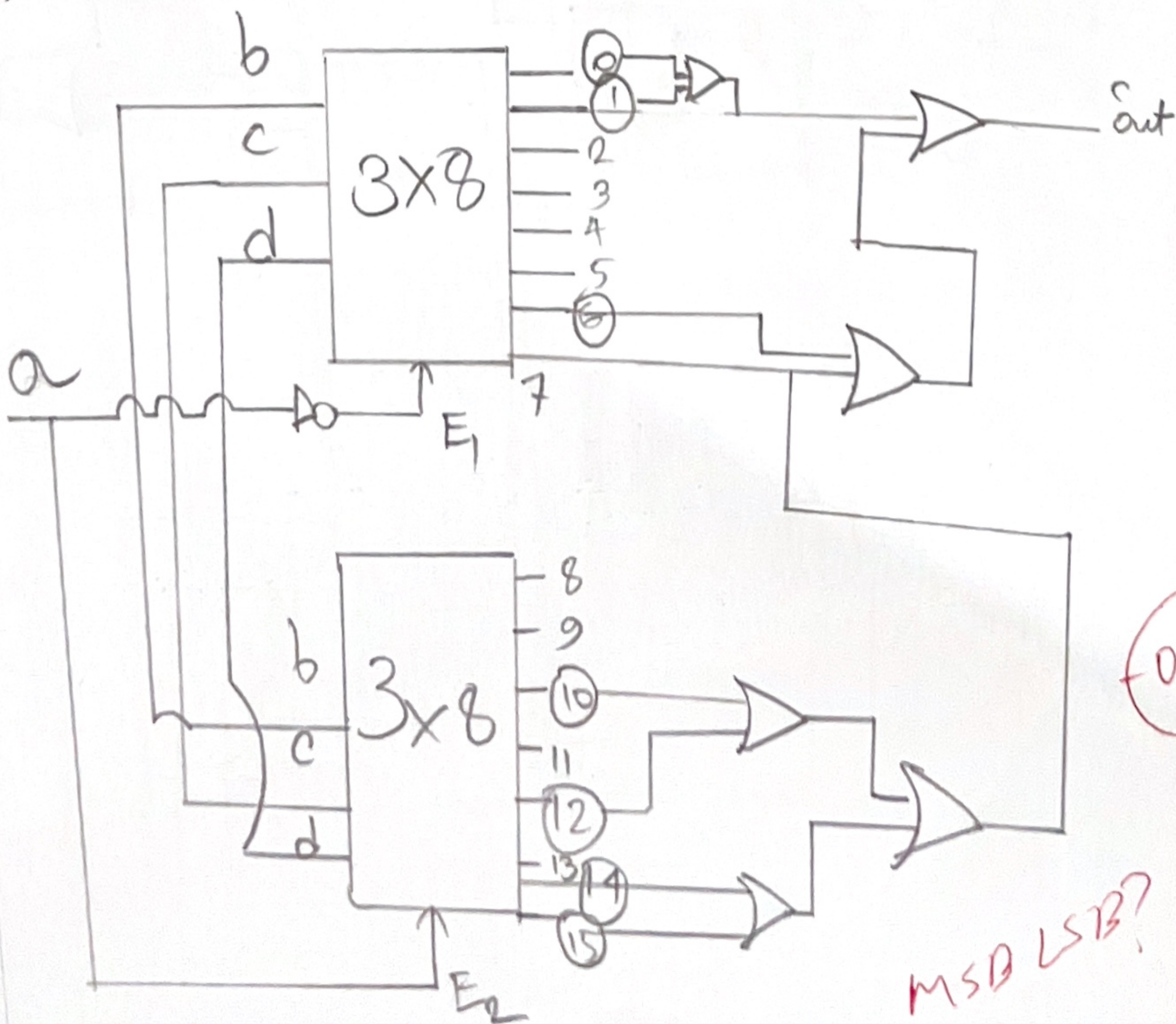
(-1)

Ans: to the Q: 05

Given,

$$f(a,b,c,d) = \sum (0, 1, 6, 10, 12, 14, 15)$$

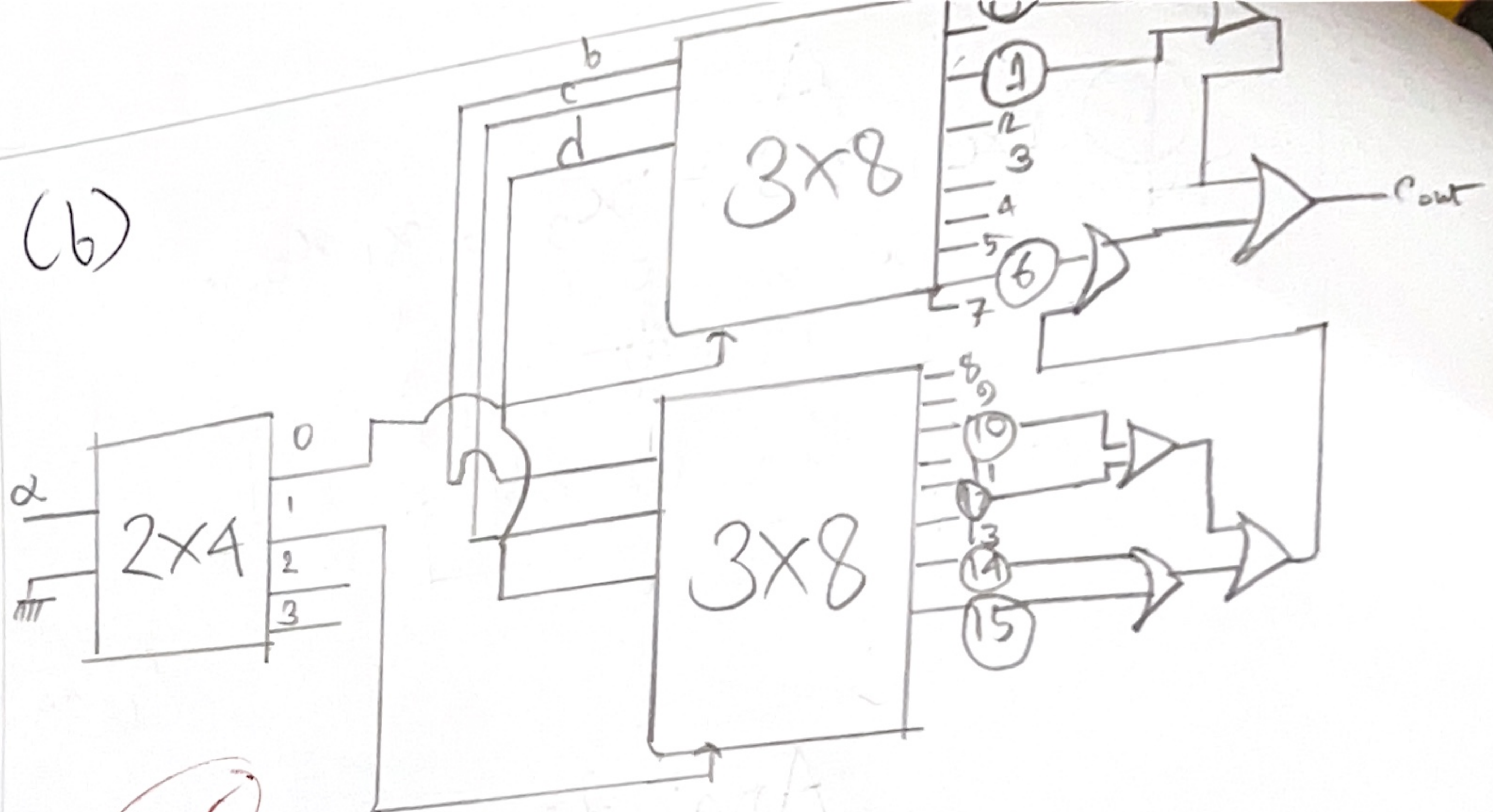
(a)



(0.5)

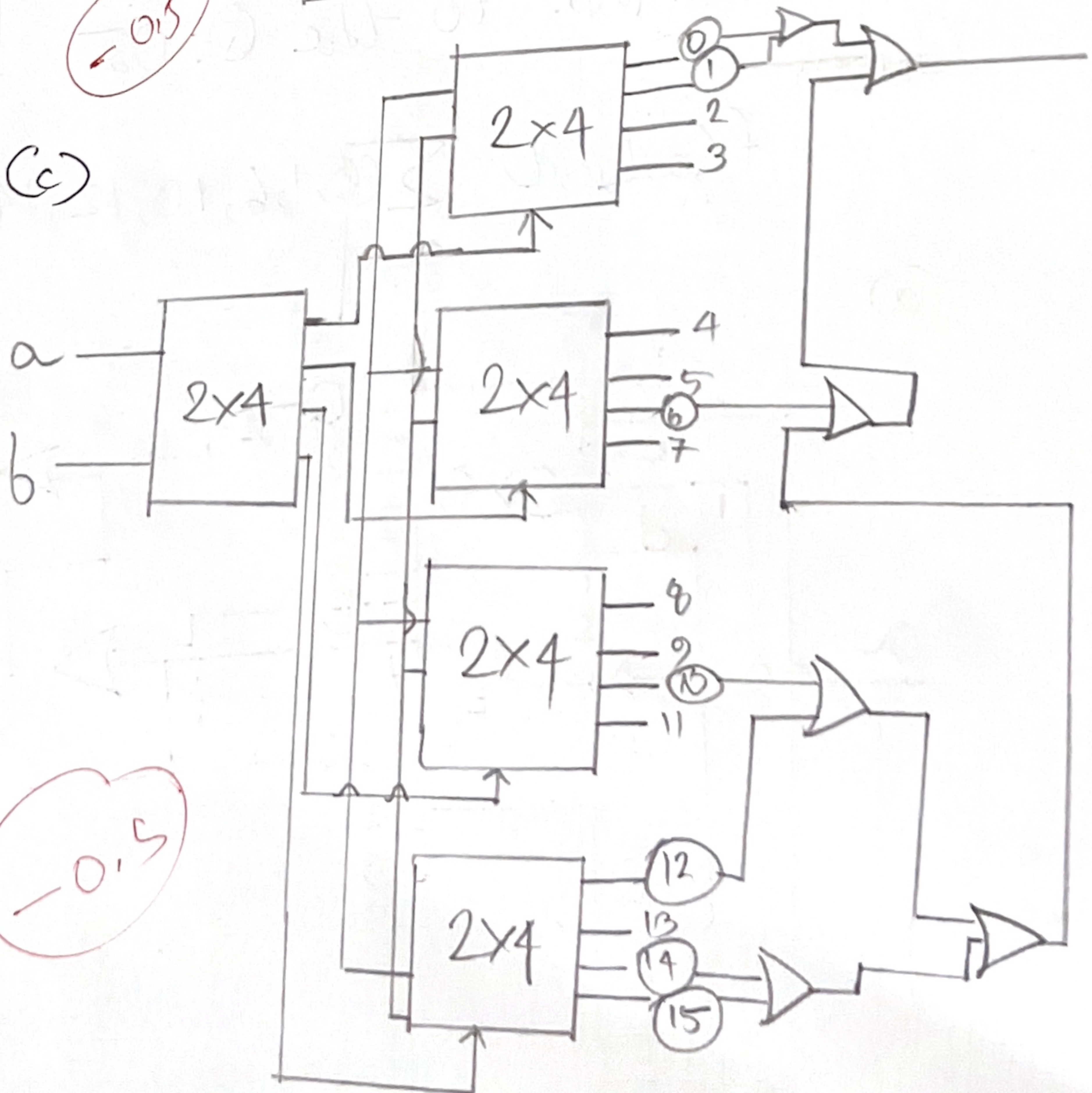
MSB LSB?

(b)



-0.5

(c)

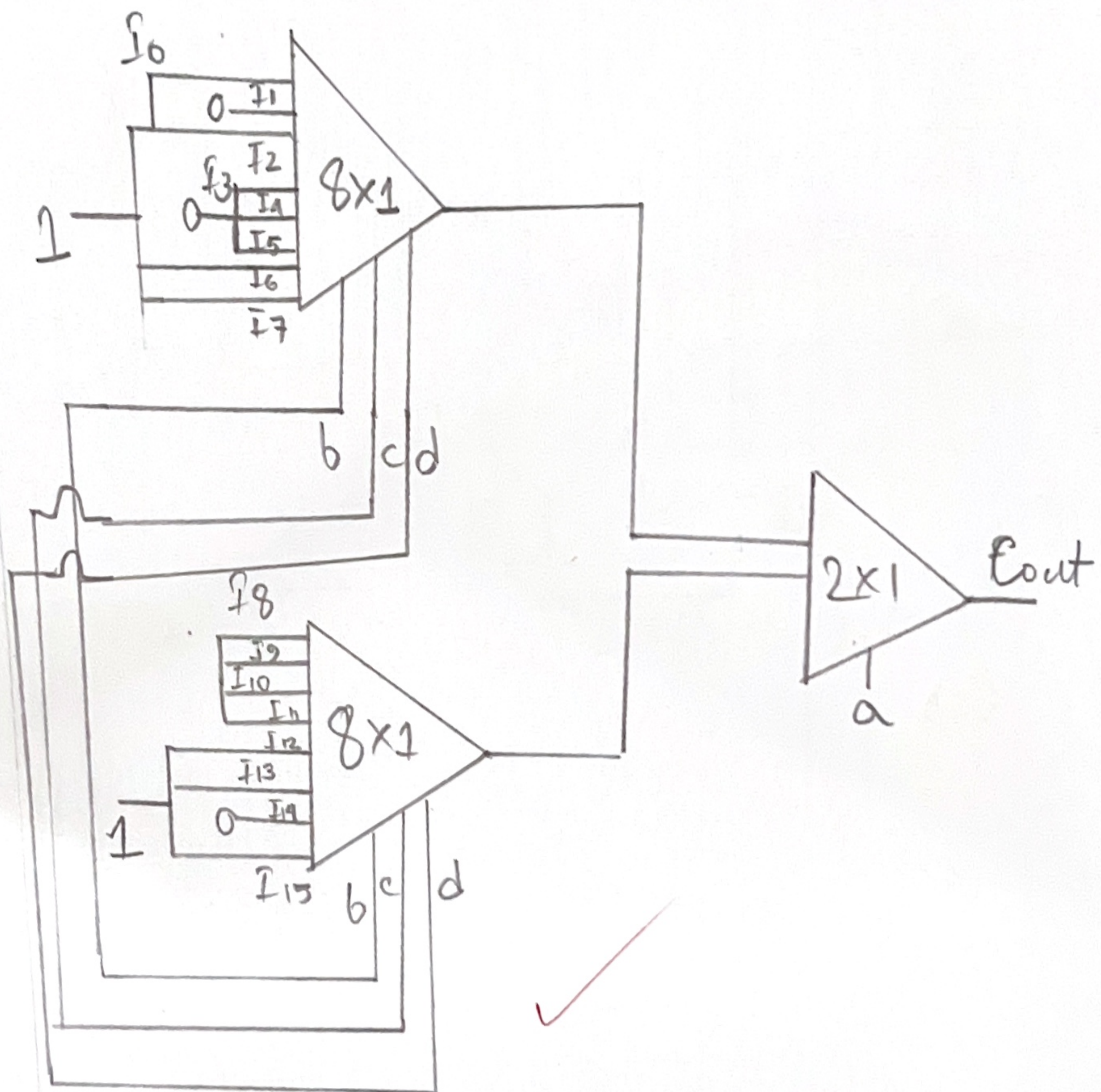


-0.5

Ans: to Q: 06

(a) Given,

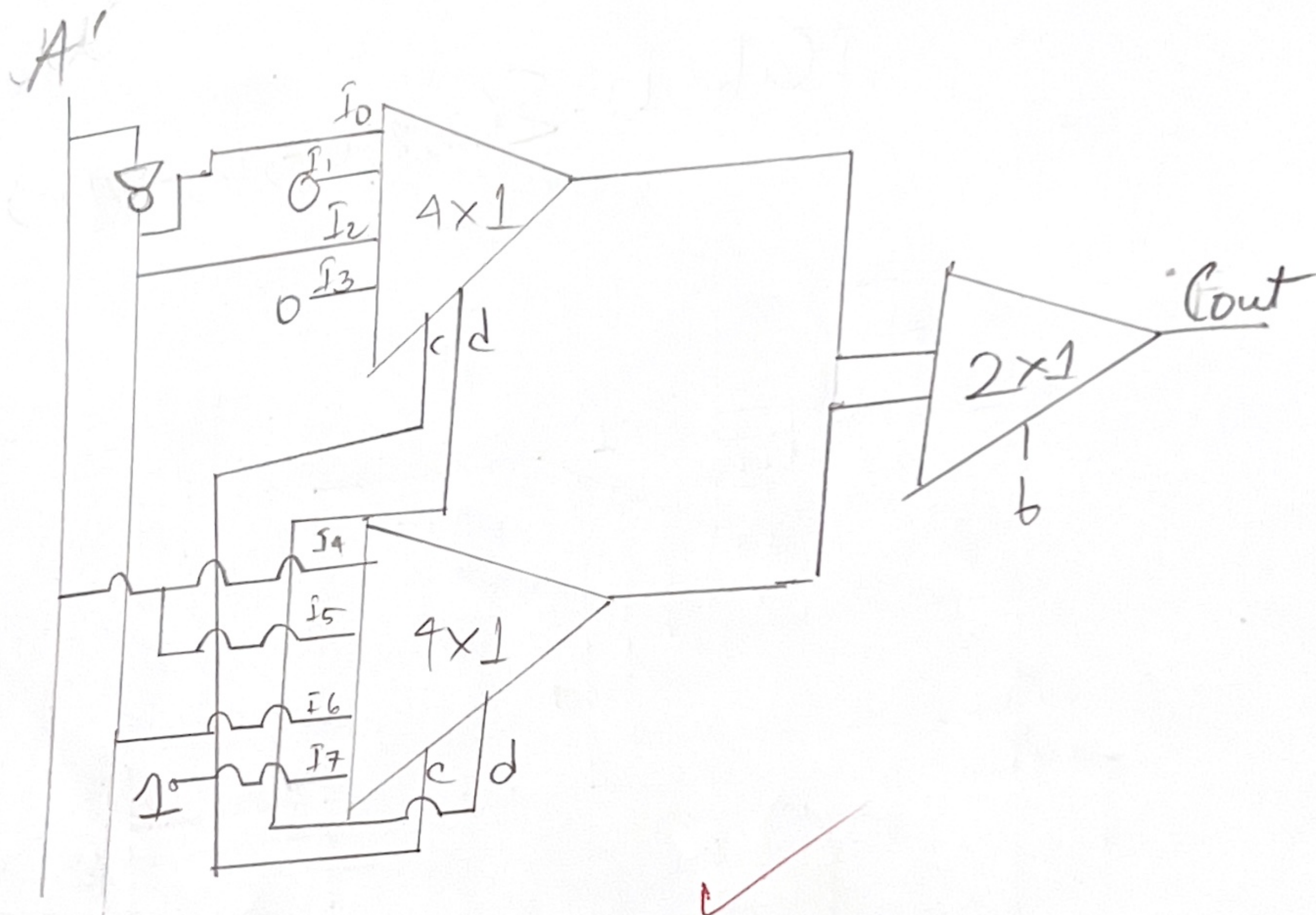
$$F(a,b,c,d) = \sum(0, 2, 6, 7, 12, 13, 15)$$



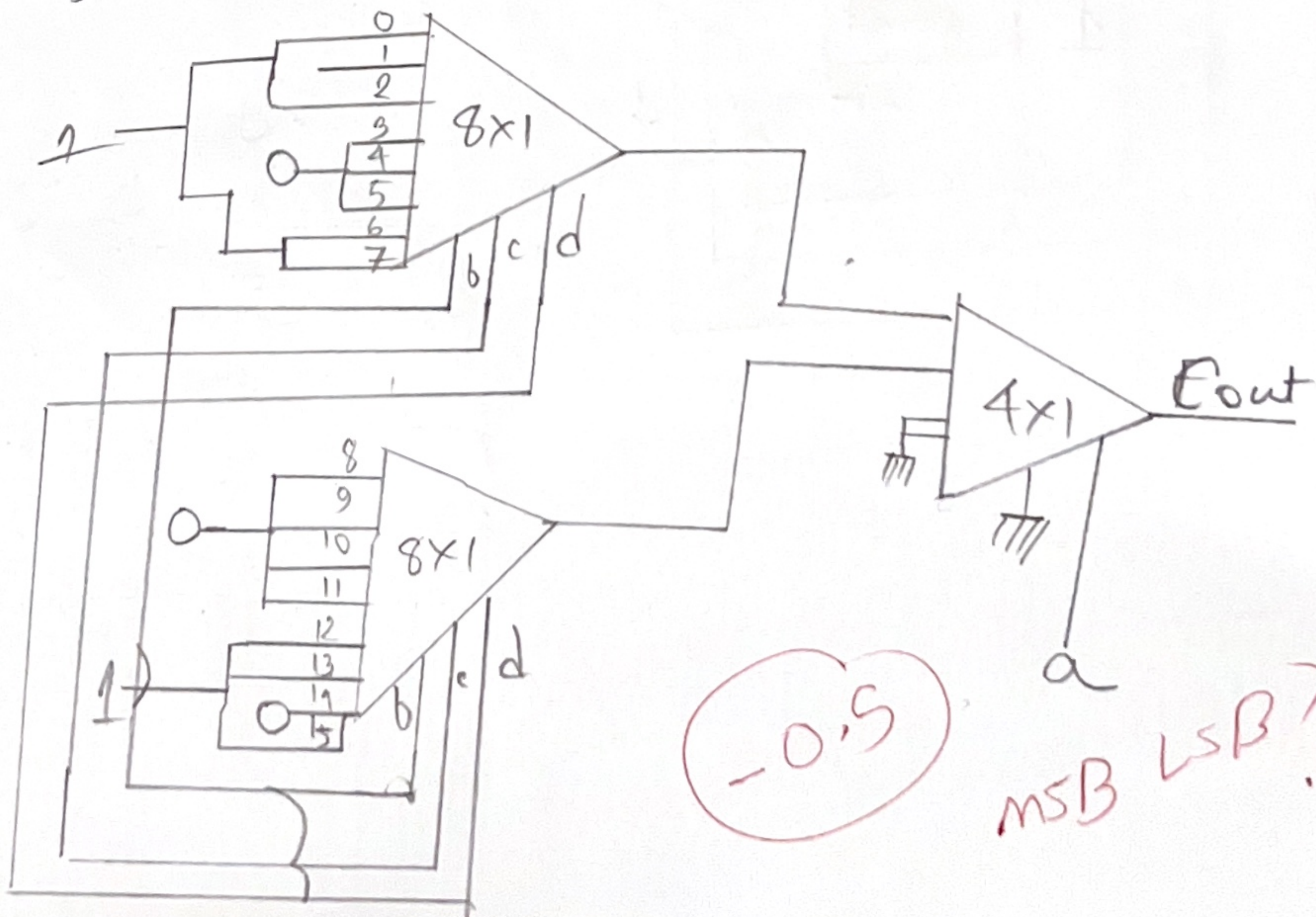
Here

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	A'	0	A'	0	A	A	A'	1

(b)



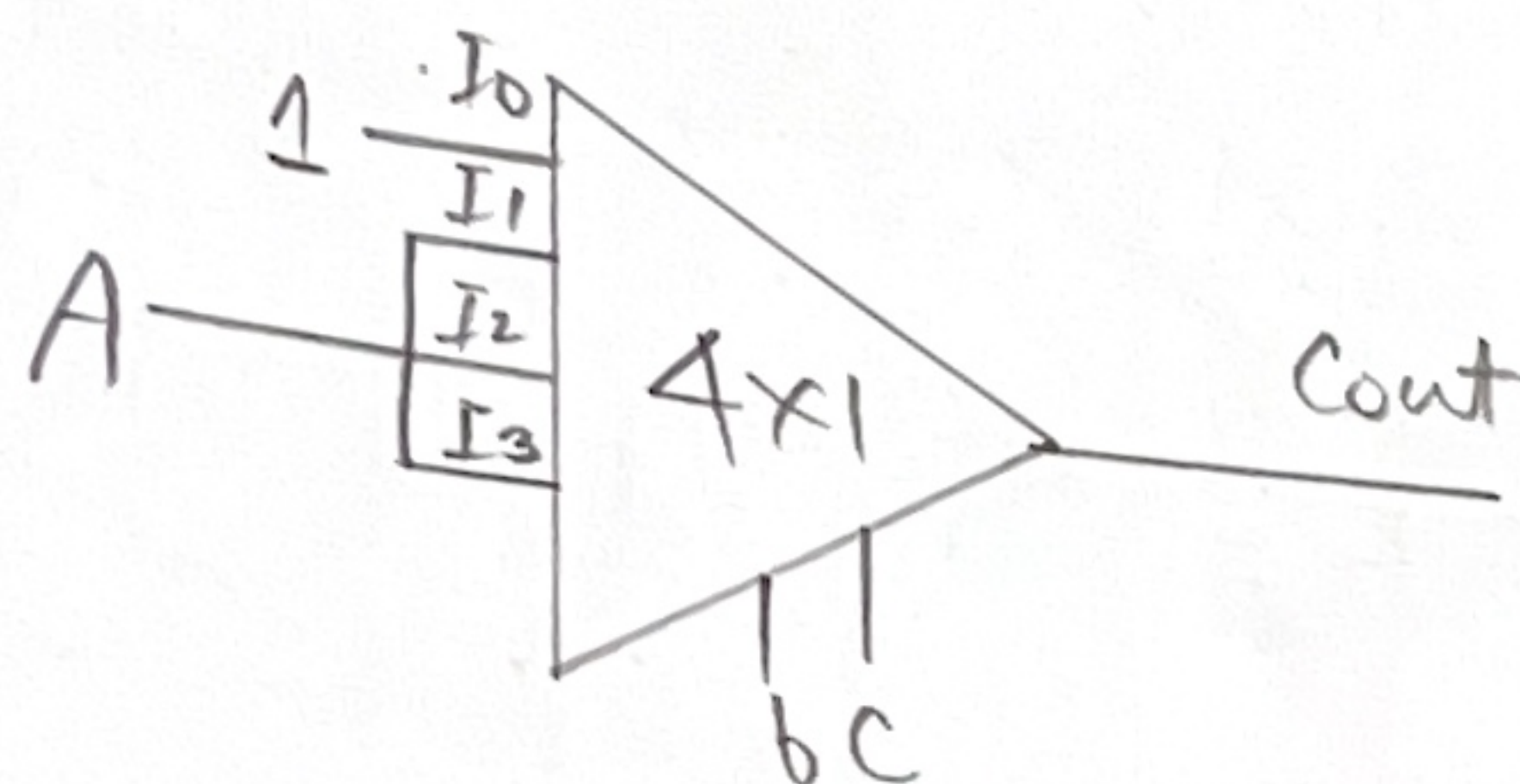
(c)



Ans: to the Q: 07

(a)

$$F(abc) = \sum (0, 4, 5, 6, 7)$$



Here,

	I_0	I_1	I_2	I_3
M'	0	1	2	3
M	4	5	6	7
	↓	↓	↓	↓
	X	M	M	M

(b) $F(abc d) = \sum (0, 2, 5, 6, 7, 9, 12, 13, 15)$

Here,

	I_0	I_1	I_2	I_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15

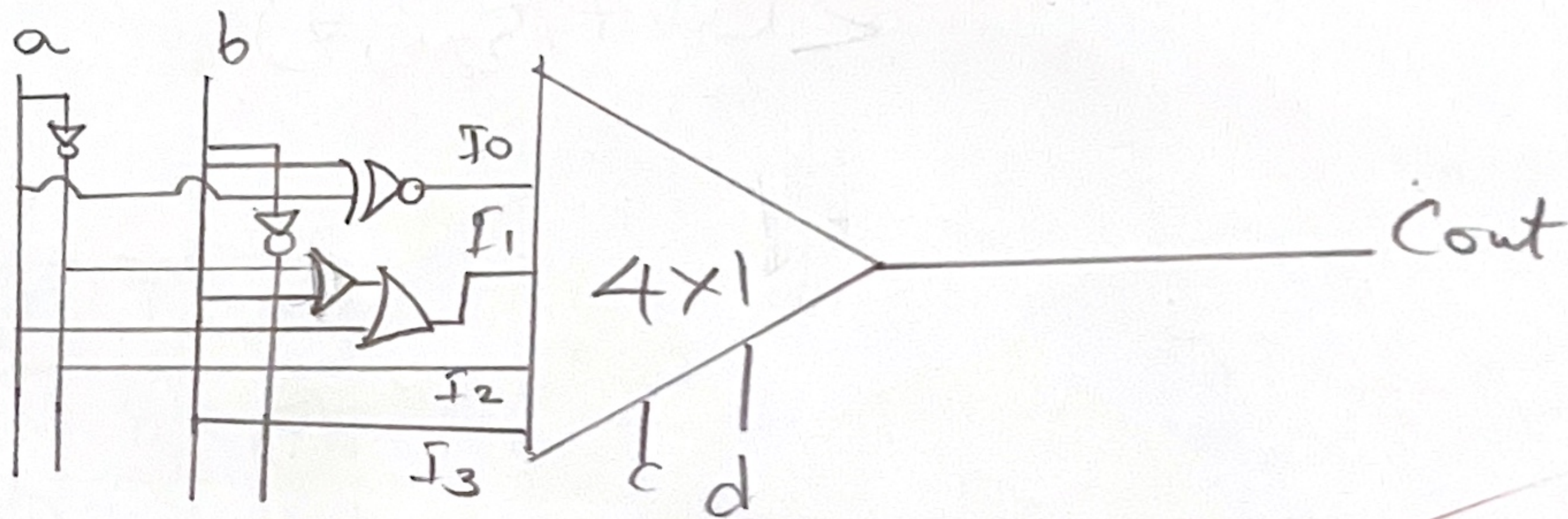
from the table:

$$I_0 = A'B' + AB \approx A \oplus B$$

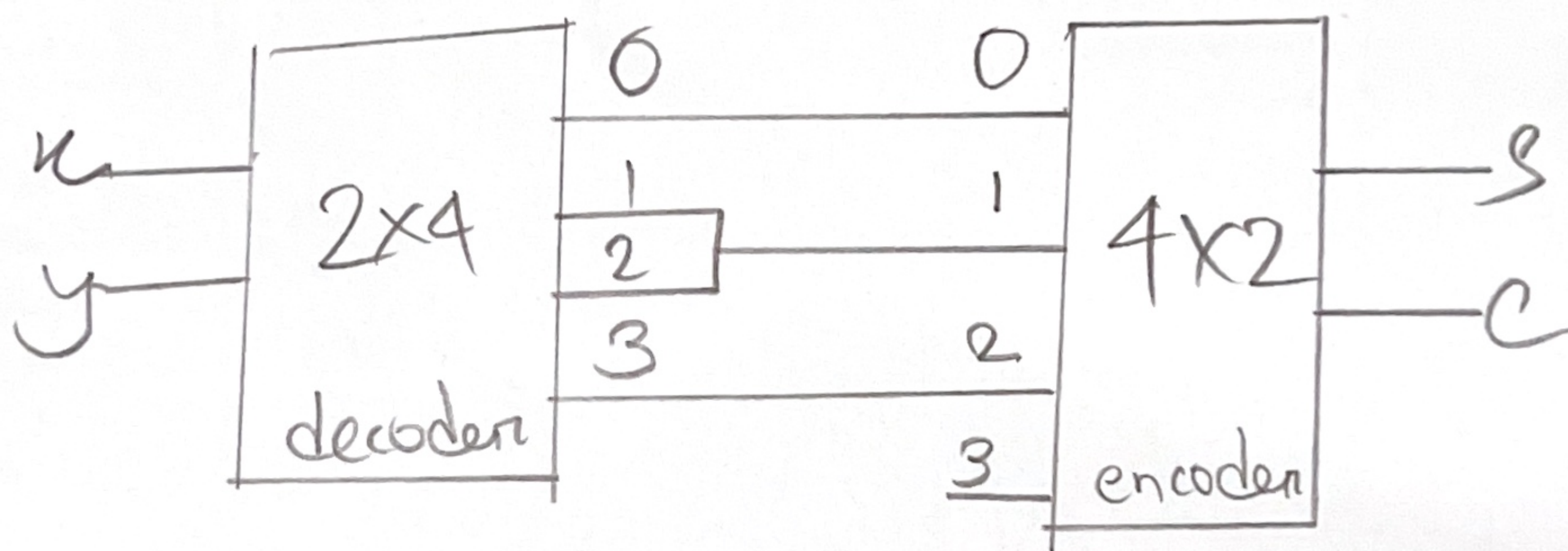
$$I_1 = A'B + AB + AB = AB + A$$

$$I_2 = A'B' + A'B = A'$$

$$I_3 = A'B + AB = B$$



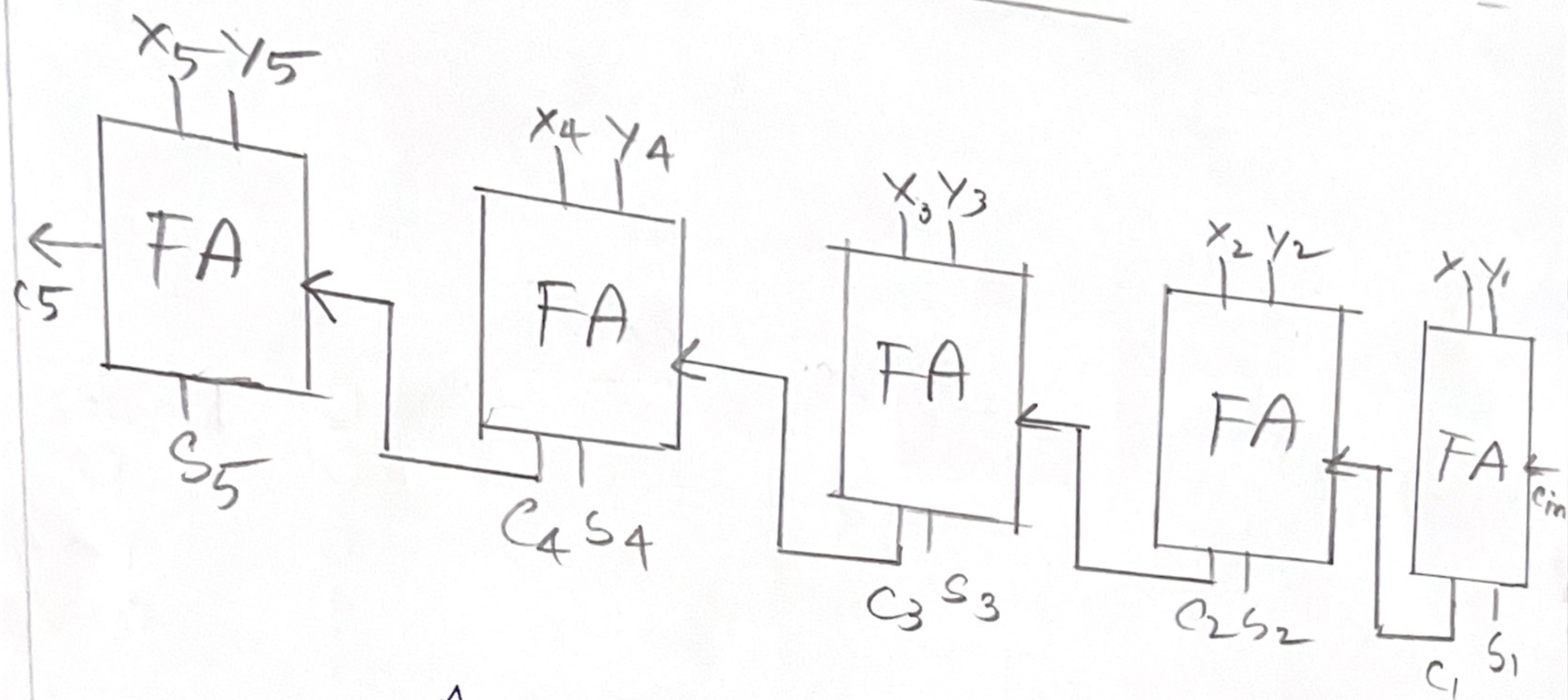
Ans: to the Q.09



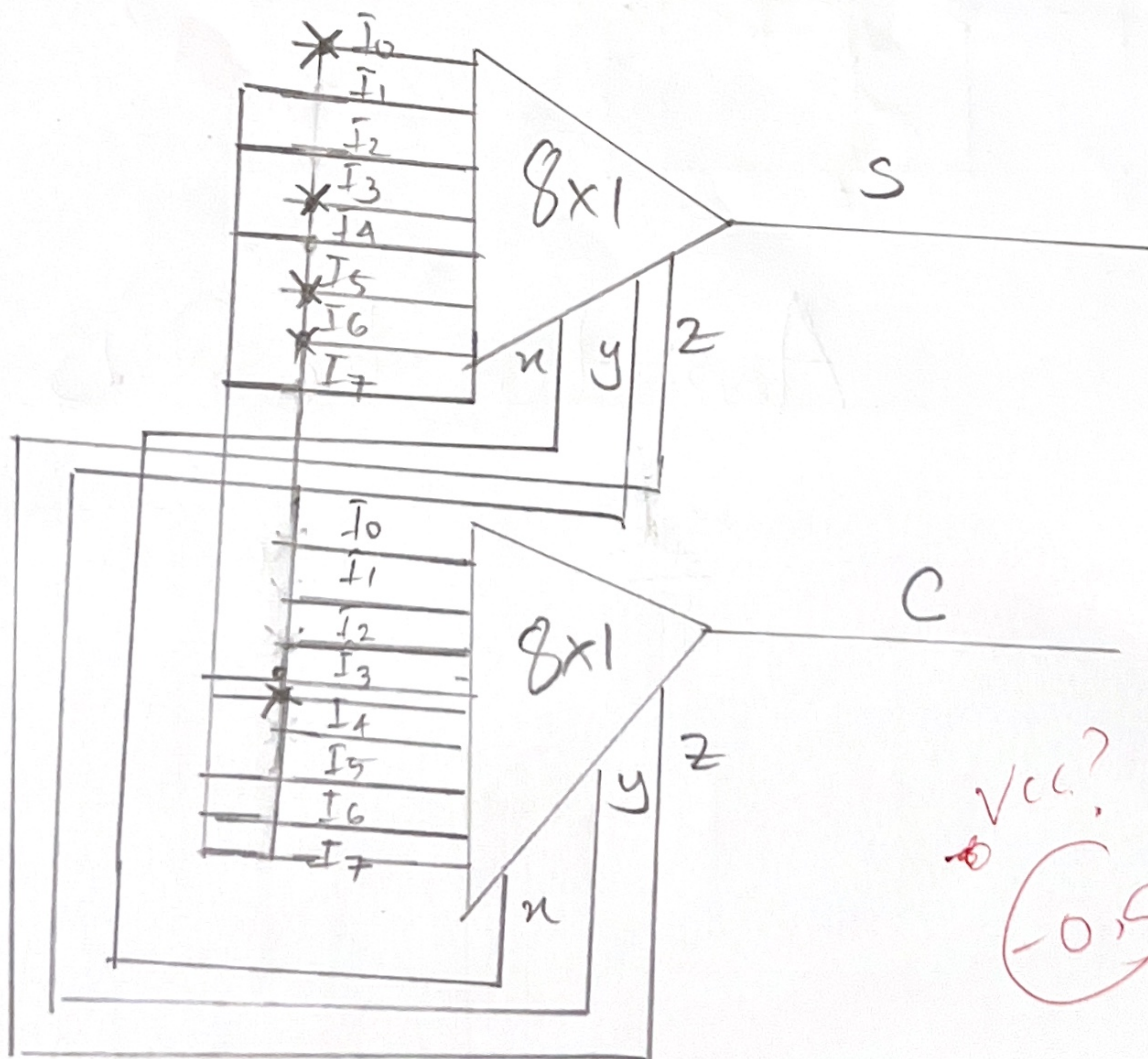
MSB
LSB?

-0.5

Ans: to Q: 10

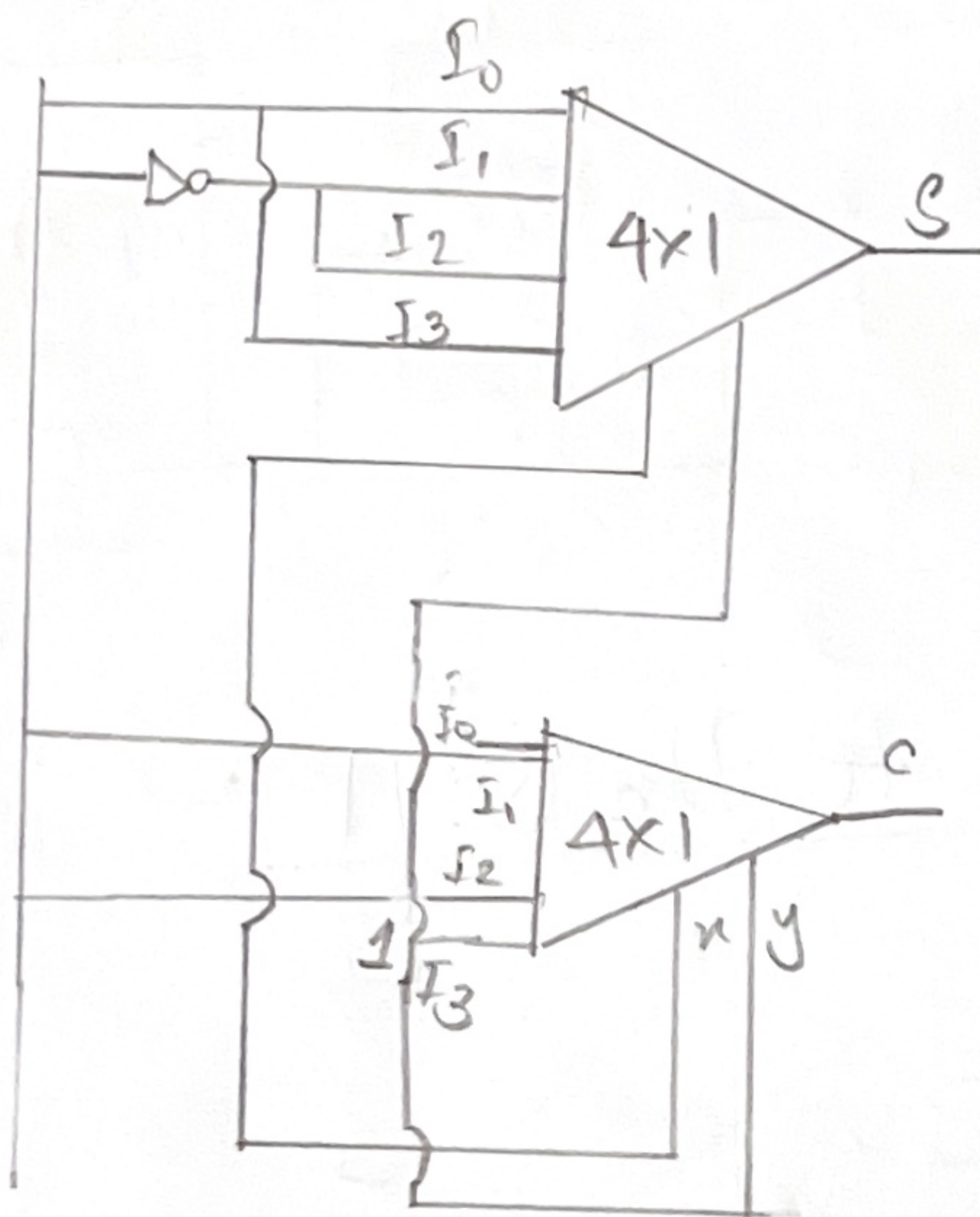


Ans: to the Q: 11 ✓



$V_{cc}?$
 -0.5

Ans: to Q:12



$\frac{C}{0001}$

$\frac{S}{0}$

$1 \rightarrow 1, 2, 4$

$0 \rightarrow 3, 5, 6$

$1 \rightarrow 7$

$-b, S$

Input?

Ans: to the Q:13

